# Multiplying and Dividing Decimals by Place-Value Reasoning
# Answer Key

Grade 4–5 Number and Operations in Base Ten

**Quick Answers**

| | | | |
|---|---|---|---|
| 1. 124 | 4. 28 | 7. — | 10. 24, 1.4 |
| 2. 0.45 | 5. 0.1 | 8. 0.8 | |
| 3. 2.4 | 6. 8 | 9. 0.49 | |

**Common wrong answers**

1. *If they got 0.0124: shifted the digits the wrong way (divided by 100 instead of multiplying)*
6. *If they got 0.125: divided the smaller number by the larger one*
9. *If they got 4.9: counted only one decimal place in the factors, so the point landed one place too far right*

**1. 100 paper clips at 1.24 g each.** Multiplying by 100 makes every digit worth 100 times as much, so each digit moves *two places to the left*: the 1 one becomes 1 hundred, the 2 tenths become 2 tens, the 4 hundredths become 4 ones.

$$1.24 \times 100 = 124$$

Sense check: 100 clips of about 1 g each should be about 100 g, and 124 g is close to that. Total mass: $\boxed{124 \text{ g}}$.

**2. 4.5 m ribbon cut into 10 equal pieces.** Dividing by 10 makes every digit worth one tenth as much, so each digit moves *one place to the right*: 4 ones become 4 tenths and 5 tenths become 5 hundredths.

$$4.5 \div 10 = 0.45$$

The answer must be smaller than 4.5 because one piece is only a tenth of the whole ribbon. Each piece: $\boxed{0.45 \text{ m}}$.

**3. 0.4 L of stock, 6 batches.** Think in tenths: $0.4 = 4$ tenths, so six batches use $6 \times 4 = 24$ tenths. Twenty-four tenths regroups as 2 ones and 4 tenths.

$$0.4 \times 6 = 2.4$$

Sense check: a bit less than half a litre per batch, six batches, so a bit less than 3 L. Stock needed: $\boxed{2.4 \text{ L}}$.

**4. Estimate the area of a 6.8 m by 4.2 m bed.** Round each measurement to the nearest whole metre: 6.8 rounds up to 7 (8 tenths is past halfway) and 4.2 rounds down to 4 (2 tenths is short of halfway).

$$7 \times 4 = 28$$

Estimated area: $\boxed{28\ \text{m}^2}$. (The exact area is 28.56 $\text{m}^2$, so the estimate is a good one — accept the estimate, not the exact value, since the problem asked for rounding.)

**5. Tile 0.25 m by 0.4 m.** Multiply the digits as whole numbers first: $25 \times 4 = 100$. Now place the point by counting place value: hundredths times tenths gives thousandths, so 25 hundredths $\times$ 4 tenths = 100 thousandths = 0.100.

$$0.25 \times 0.4 = 0.1$$

On the hundredths grid the tile covers one tenth of the square metre — a small part of one square metre, which matches multiplying two numbers that are each less than 1. Area: $\boxed{0.1\ \text{m}^2}$.

**6. 4.8 m rope cut into 0.6 m pieces.** Rewrite both lengths in the same place-value unit: $4.8 = 48$ tenths and $0.6 = 6$ tenths. Asking how many 0.6 m pieces fit in 4.8 m is asking how many 6 tenths fit in 48 tenths.

$$48 \div 6 = 8, \qquad \text{so } 4.8 \div 0.6 = 8$$

Check by multiplying back: $8 \times 0.6 = 4.8$. Number of pieces: $\boxed{8}$.

**7. Compare** $0.6 \times 12$ **and** $12 \div 0.6$ **without computing.** Multiplying 12 by a factor smaller than one makes it *smaller* than 12. Dividing 12 by a number smaller than one asks how many small pieces fit inside 12, and many of them fit, so the quotient is *larger* than 12. One value is below 12 and the other is above 12, so the correct symbol is $\boxed{<}$.
For confirmation only (the student was not asked to compute): $0.6 \times 12 = 7.2$ and $12 \div 0.6 = 20$.

**8. Nine books, 7.2 kg in total.** Think in tenths: $7.2 = 72$ tenths, and 72 tenths shared into 9 equal groups gives $72 \div 9 = 8$ tenths in each group.

$$7.2 \div 9 = 0.8$$

Check: $9 \times 0.8 = 7.2$. Each book: $\boxed{0.8\ \text{kg}}$.

**9. Find the mistake: 1.4 m by 0.35 m.** Estimate first: 1.4 is about 1 and 0.35 is about a third, so the area must be well under one square metre. Sam's 4.9 is bigger than either side length, which is impossible when one factor is less than 1 — his decimal point is one place too far right. Correct work: $14 \times 35 = 490$, and one decimal place times two decimal places gives three decimal places, so 0.490.

$$1.4 \times 0.35 = 0.49$$

Correct area: $\boxed{0.49\ \text{m}^2}$.

**10. Juice machine: 8.4 L, cups of 0.35 L.** (a) Rewrite both amounts in hundredths: $8.4 = 840$ hundredths and $0.35 = 35$ hundredths. Then $840 \div 35 = 24$, so

$$8.4 \div 0.35 = 24$$

Check: $24 \times 0.35 = 8.4$. Full cups: $\boxed{24}$.
(b) Twenty cups use $20 \times 0.35 = 7.0$ litres, so what is left is

$$8.4 - 7 = 1.4$$

Juice remaining: $\boxed{1.4\text{ L}}$.